1. Loss Function for Multiclass Classification:

A. error function $= e =$ Mean Square Error (MSE) $= \frac{1}{m}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}(y_{ij} - \hat{y}_{ij})^2$,

y_{ij} = desire output y (ground truth) from i-th pair training example which corresponding to j-th output perceptron

$\hat{y}_{ij}$ = predicted output P_y from j-th output perceptron which corresponding to i-th pair training example

m = number of pair training examples

n = number of output perceptrons in the neural network.

B. Overall Error of j-th output perceptron:

$\Rightarrow$ e individually partial derivate $\sum_{i=0}^{m-1}\hat{y}_{ij}$ $(= \hat{y}_{0j}, \hat{y}_{1j}, \hat{y}_{2j}, \ldots\ldots \hat{y}_{(m-1)j})$

$$\Rightarrow \sum_{i=0}^{m-1} e'(\hat{y}_{ij}) = e'(\hat{y}_{0j}) + e'(\hat{y}_{1j}) + e'(\hat{y}_{2j}) + \ldots\ldots e'(\hat{y}_{(m-1)j})$$

$$\Rightarrow \sum_{i=0}^{m-1}\frac{\partial e}{\partial \hat{y}_{ij}} = \frac{\partial e}{\partial \hat{y}_{0j}} + \frac{\partial e}{\partial \hat{y}_{1j}} + \frac{\partial e}{\partial \hat{y}_{2j}} + \ldots\ldots \frac{\partial e}{\partial \hat{y}_{(m-1)j}}$$

$$\Rightarrow -\frac{2}{m}\sum_{i=0}^{m-1}(y_{ij} - \hat{y}_{ij}) = -\frac{2}{m}(y_{0j} - \hat{y}_{0j}) + -\frac{2}{m}(y_{1j} - \hat{y}_{1j}) + -\frac{2}{m}(y_{2j} - \hat{y}_{2j}) + \ldots\ldots + -\frac{2}{m}(y_{(m-1)j} - \hat{y}_{(m-1)j})$$

$$\Rightarrow \text{Overall Error}_{j\text{-th output perceptron}} = -\frac{2}{m}\sum_{i=0}^{m-1}(y_{ij} - \hat{y}_{ij})$$